

IK 539: Sanskrit and Technology: An Overview

Course Code	IK 539
Course Name	Sanskrit and Technology: An Overview
L-T-P-C	3-0-0-3
Prerequisites	IK 540 Bhagavad-Gītā Part II or equivalent understanding of chapters 7-12.
Intended for	BTech/MTech/MS/MSc/MA/Ph.D
Mutual Exclusion	None
Approval	42nd Senate (Jan 2024)

Course Contents

- **Interactive Sanskrit Teaching Learning Tools:** (12 hours)
 - Introduction to Interactive Sanskrit Learning Tools, Why Interactive Tools for Sanskrit?
 - * E-learning, Basics of Multimedia, Web-based tools development, HTML, Web page, etc., Tools and Techniques
- **Standard for Indian Languages (Unicode)** (9 hours)
 - Nature of Devanagari/Brahmni scripts, Concept of Aksharas, Conjuncts, and Script grammar.
 - * Typing in Devanagari Scripts, Typing Tools and Software
- **Text Processing and Preservation Tools** (12 hours)
 - Text Processing, Preservation Techniques, Text Processing and Preservation Tools, and
 - * Techniques, Survey
- **Optical Character Reader:** Word generation, word analysis, compound formation, sandhi generation and splitting, sentence analysis, Optical Character Reader (OCR), Applications of OCR for Sanskrit and Indian Languages, Tool and Techniques, Survey. (12 hours)

Text Books

1. NA

Readings

1. Teacher's notes, ppt, and handout
2. Bharti A., R. Sangal, V. Chaitanya, "NL, Complexity Theory and Logic" in Foundations of Software Technology and Theoretical Computer Science, Springer, 1990.
3. E-Content suggested by Teacher
4. Tools developed by Computational Linguistics Group, Department of Sanskrit, University of Delhi, Delhi-110007 available at: <http://sanskrit.du.ac.in>
5. Basic concept and issues of multimedia: <http://www.newagepublishers.com/samplechapter/001697.pdf>
6. Content creation and E-learning in Indian languages: a model: http://eprints.rclis.org/7189/1/vijayakumar_jk_01.pdf
7. HTML Tutorial - W3Schools:
8. The Unicode Consortium: <http://unicode.org/>